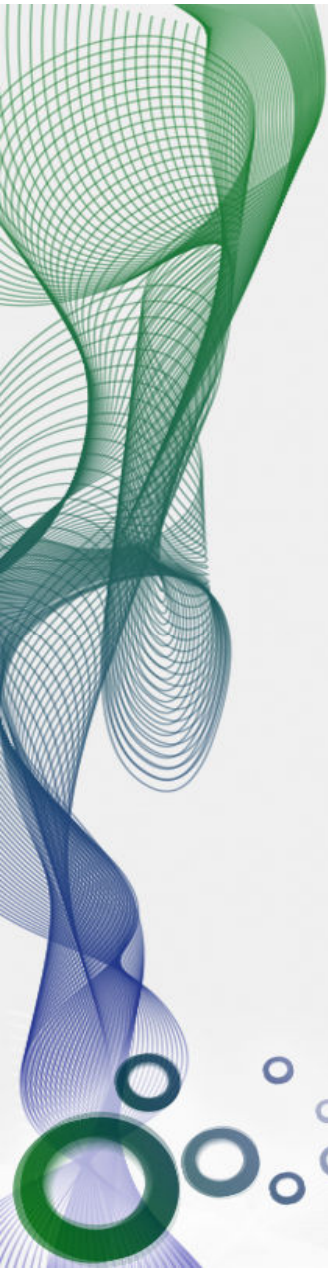
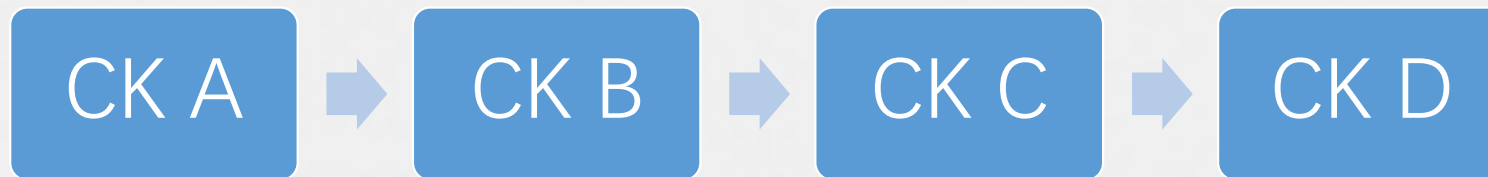


Assay for Enzyme Activity



Creatine Kinase Purification



A. Preparation & Separation

- Crude extract prep – salting in
- Ammonium sulfate precipitation
- Ethanol precipitation
- Fractional extraction – salting out

C. Enzyme activity measurement

- Bradford assay
- Specific activity

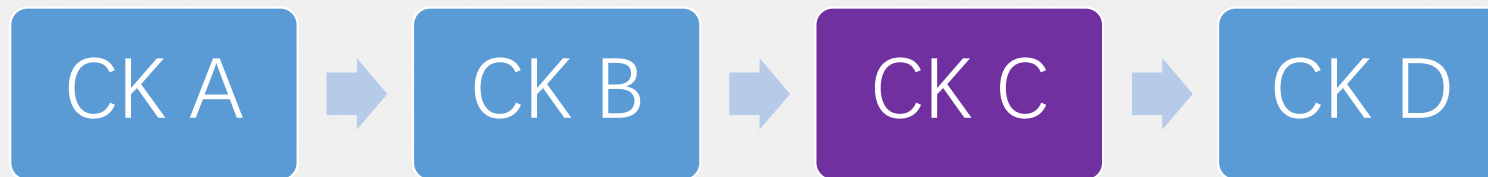
B. Purification through Chromatography

- Desalting-Gel filtration
- Anion exchange chromatography

D. Electrophoresis

- SDS-PAGE
- WB

Creatine Kinase Purification



A. Preparation & Separation

- Crude extract prep – salting in
- Ammonium sulfate precipitation
- Ethanol precipitation
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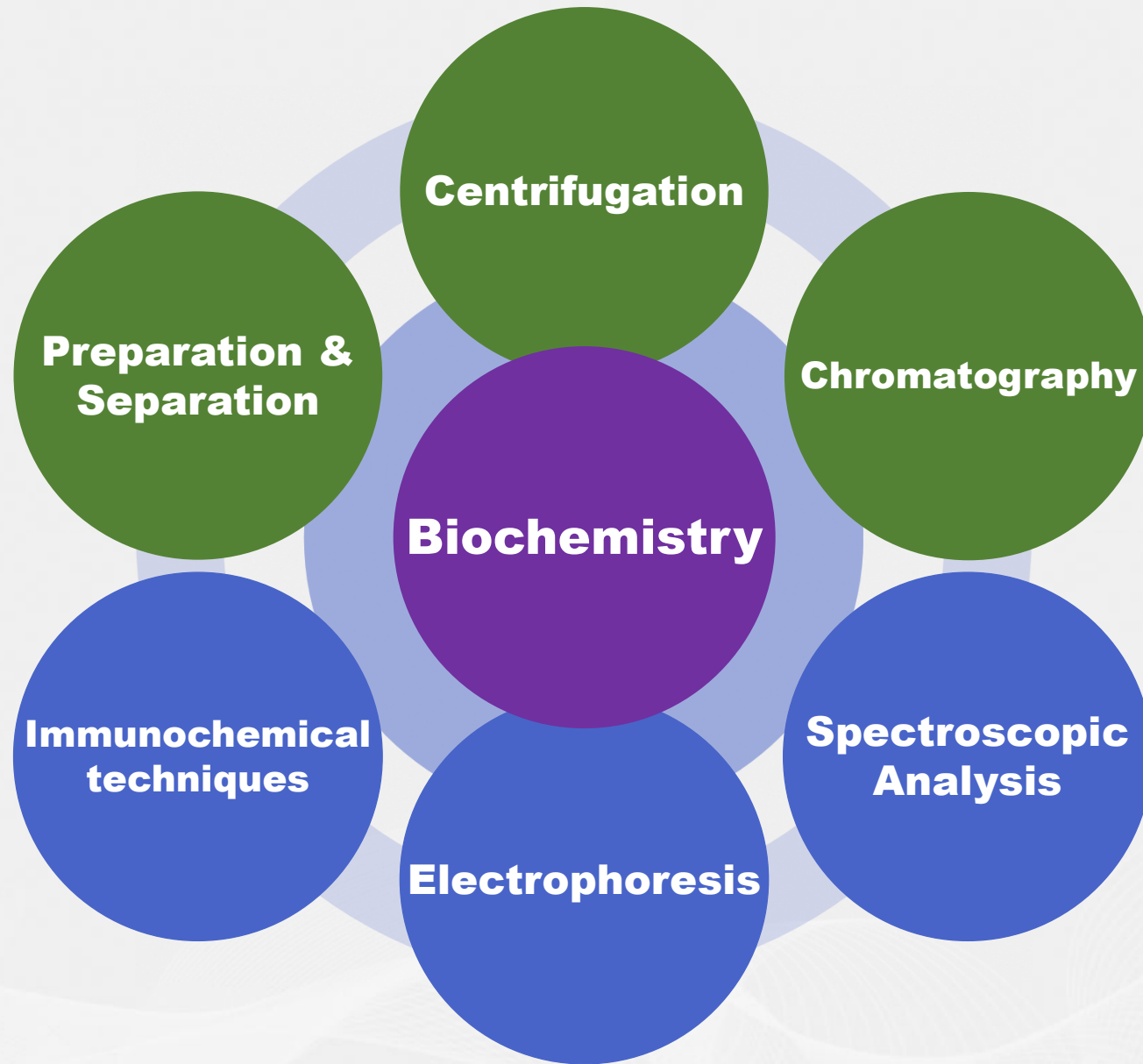
D. Electrophoresis

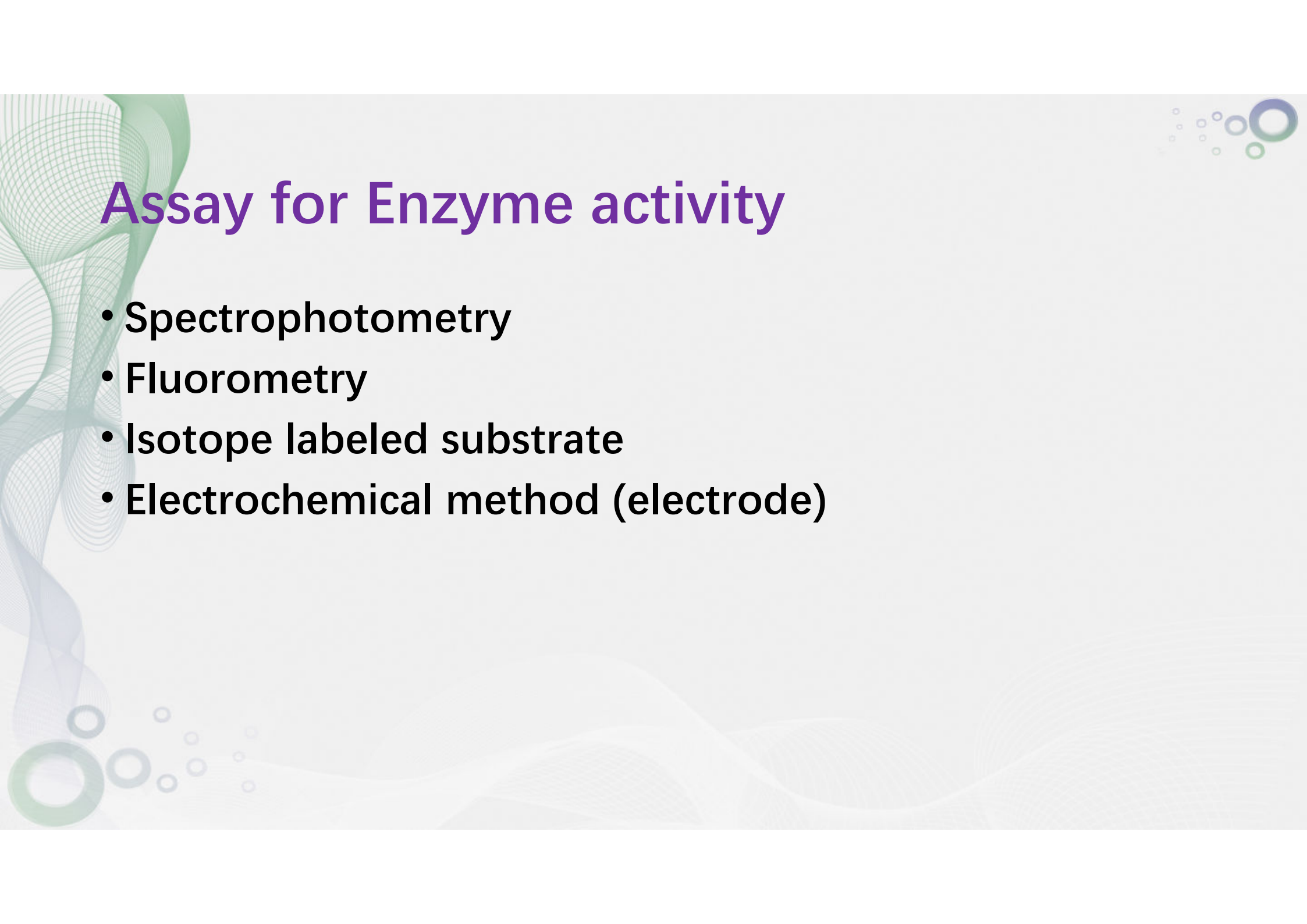
- SDS-PAGE
- WB



Rabbit Muscle Creatine Kinase (CK) Purification

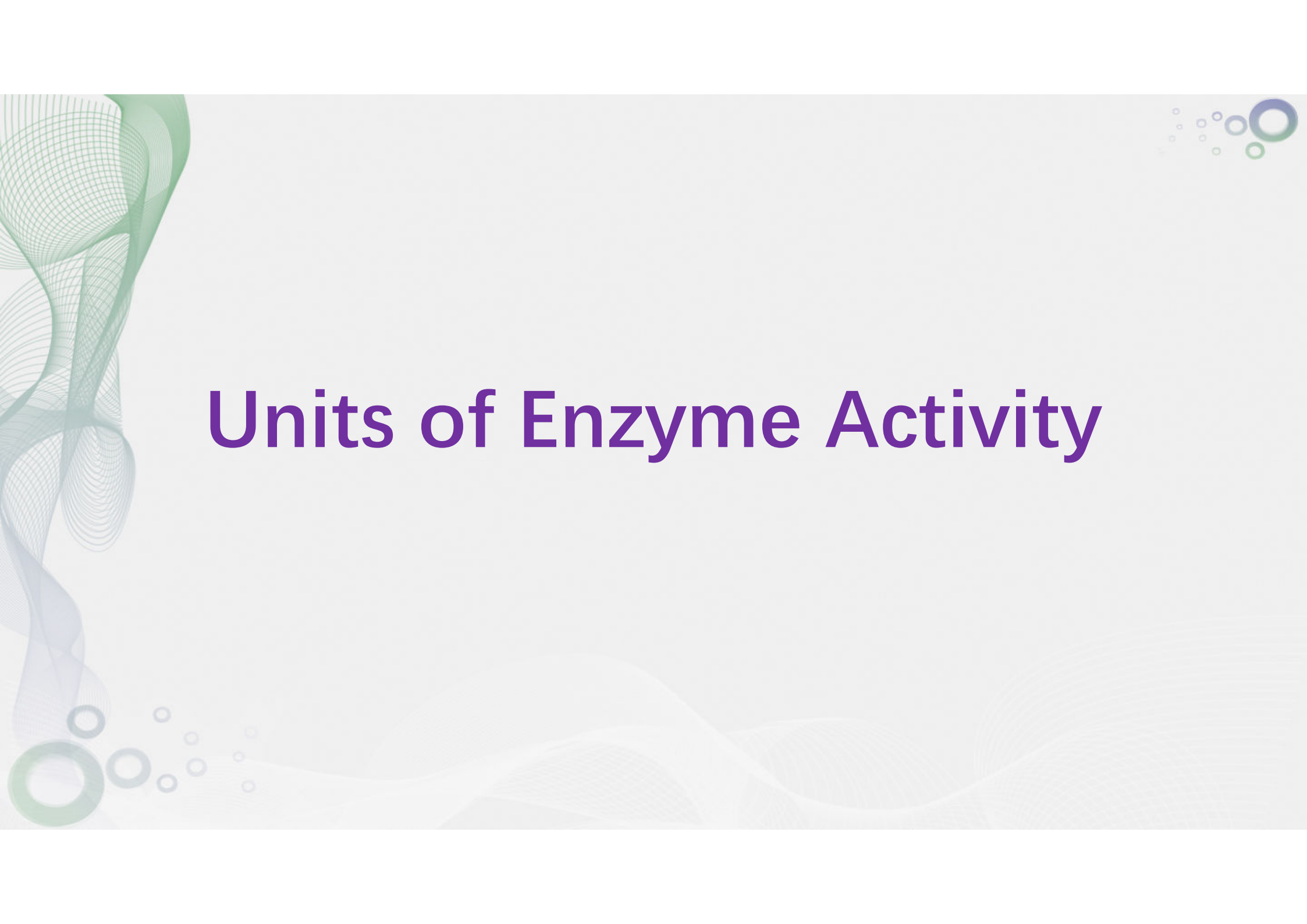
- Homogenization / Salting in / Precipitation (pI / salting out / organic solution)
- Centrifugation
- SDS-PAGE
- IEx Chromatography
- Colorimetric method of CK assay
- Western blot





Assay for Enzyme activity

- Spectrophotometry
- Fluorometry
- Isotope labeled substrate
- Electrochemical method (electrode)



Units of Enzyme Activity

Units of Enzyme Activity

- Activity of enzyme - **what those molecules are doing**
- Activity ~ amount of product/time
- 1 unit of activity (U) = $1\text{ }\mu\text{mol/min}$ (国际单位 IU)
 - 1 unit of activity (U) = 1 nmol/min - low activity enzyme
- Reaction conditions-pH, ionic strength, temperature, present of inhibitors or activators

Units of Enzyme Activity

- Relative activity=units/mL=U/mL (U=1 μ mol/min)
- **Specific activity=U/mg protein**
- % recovery=total units of fraction/total units of crude
- Fold purification=specific activity of fraction/specific activity of crude

TABLE 3-5 Purification Table for a Hypothetical Enzyme

Procedure or step	Fraction volume (ml)	Total protein (mg)	Activity (units)	Specific activity (units/mg)
1. Crude cellular extract	1,400	10,000	100,000	10
2. Precipitation with ammonium sulfate	280	3,000	96,000	32
3. Ion-exchange chromatography	90	400	80,000	200
4. Size-exclusion chromatography	80	100	60,000	600
5. Affinity chromatography	6	3	45,000	15,000
Note: All data represent the status of the sample after the designated procedure has been carried out. Activity and specific activity are defined on page 95. After step 5, the enzyme in question has been purified by a factor of 1,500, as reflected in the increase in specific activity relative to that in the crude extract, and the yield of the enzyme is 45%, as reflected in the recovery of total activity.				



Colorimetric method of CK assay

Calculating Initial Velocity

- $\text{ATP} + \text{Cr} \rightleftharpoons \text{ADP} + \text{PCr} + \text{H}^+$
- $\text{H}^+ \Rightarrow \Delta A_{597}/\text{min}$
- Cumulative protons change the color of substrate solution (containing **thymol blue**), which absorbs maximally at 597 nm at pH 9.0, from purple to yellow, leading the read of A_{597} decreased.
- Reaction rate is linear for the first minute but then decreases.

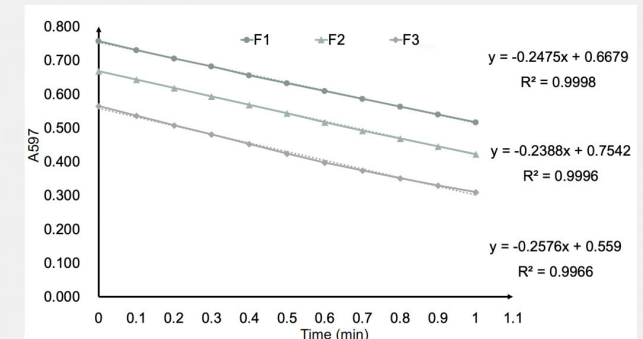


Figure 1. CK activity measured at the wavelength of 597 nm

Colorimetric method of CK assay

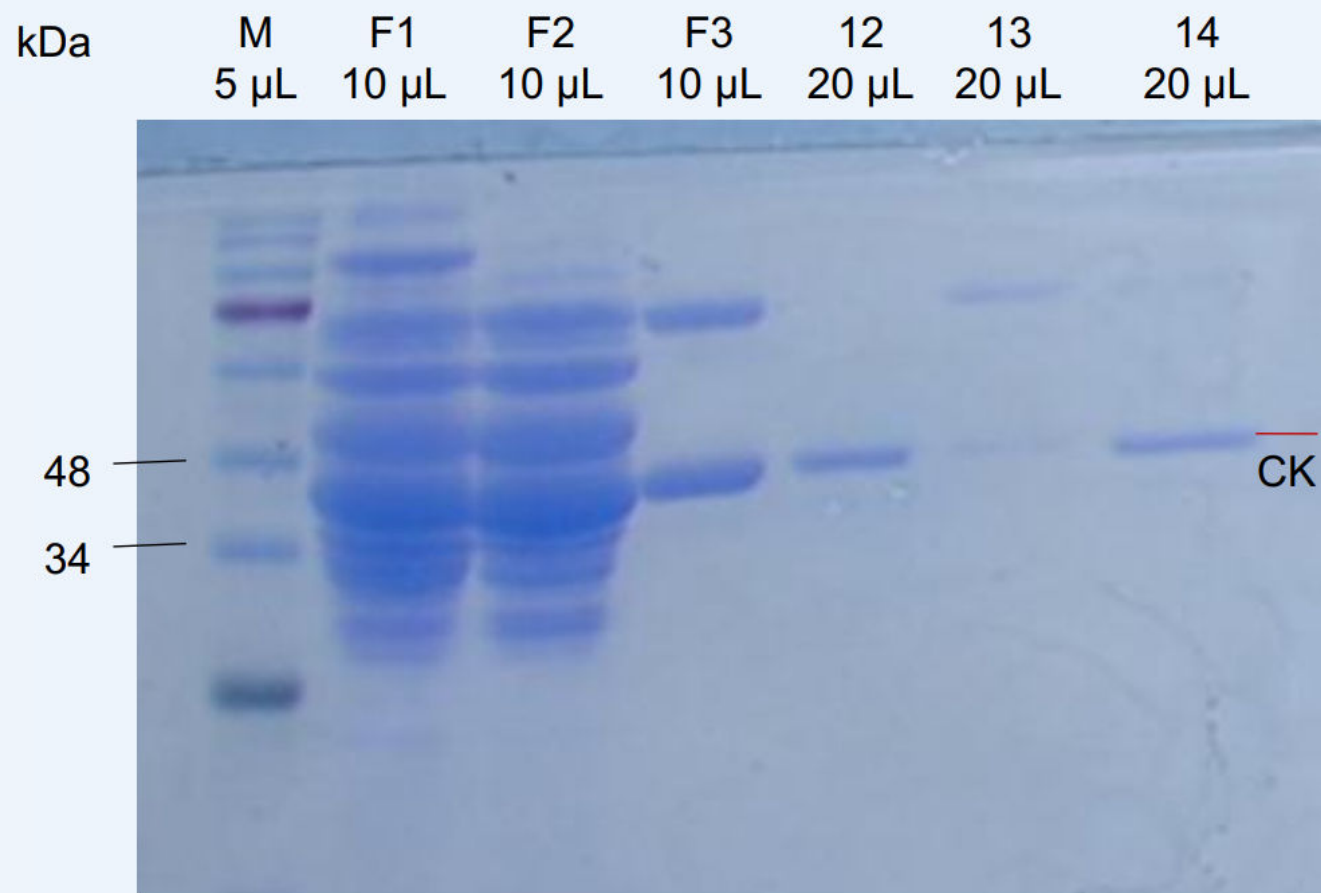
- Colorimetric method (Spectrophotometry)
 - Blanking before measurement
 - Measure absorbance change every 15 seconds / 8-12 times
 - $\Delta A_{597}/\text{min}$
- Graph the data and take the initial slope of the line.

$$U [\mu\text{mol}/(\text{min}\cdot\text{mg})] = [1.3 \times \Delta A_{597}/\text{min} \times V_A / (C \times V_B)] \times \text{dilution factor}$$

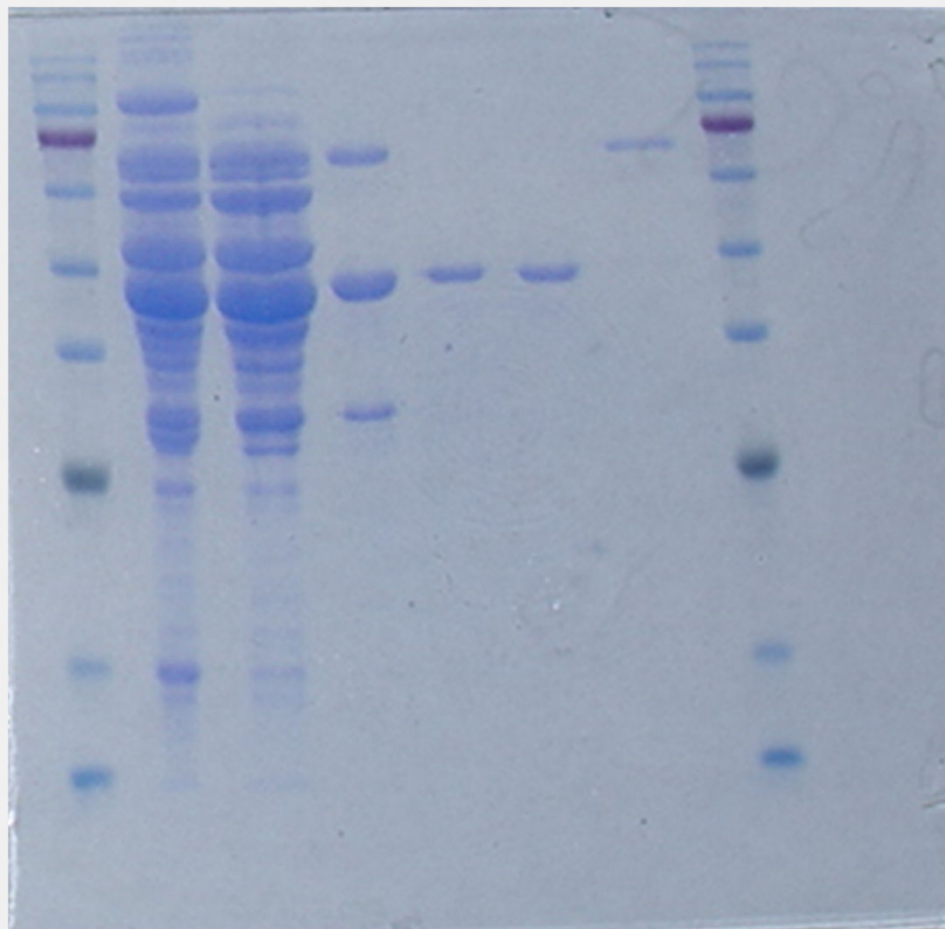
- Dilution(ddH₂O): **F1/F2-1:20-100 F3-1:10-50 ice-bath**

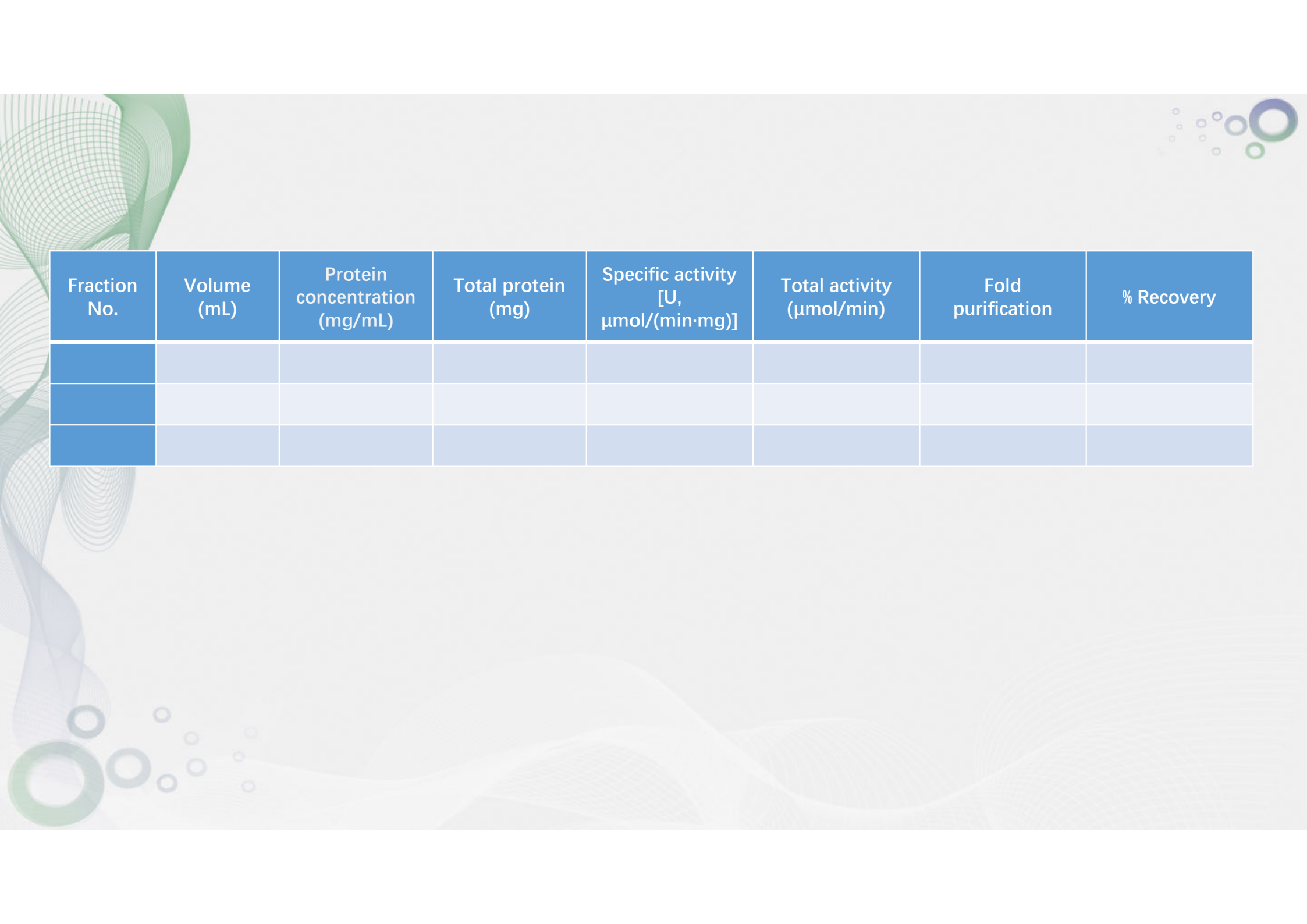
Colorimetric method of CK assay

- Take out 10 μL every time; **do not** dilute the whole stock directly; **do not** perform serial dilution.
- Keep substrate solution at room temperature
- Operation of adding enzyme?
- Wash cuvette



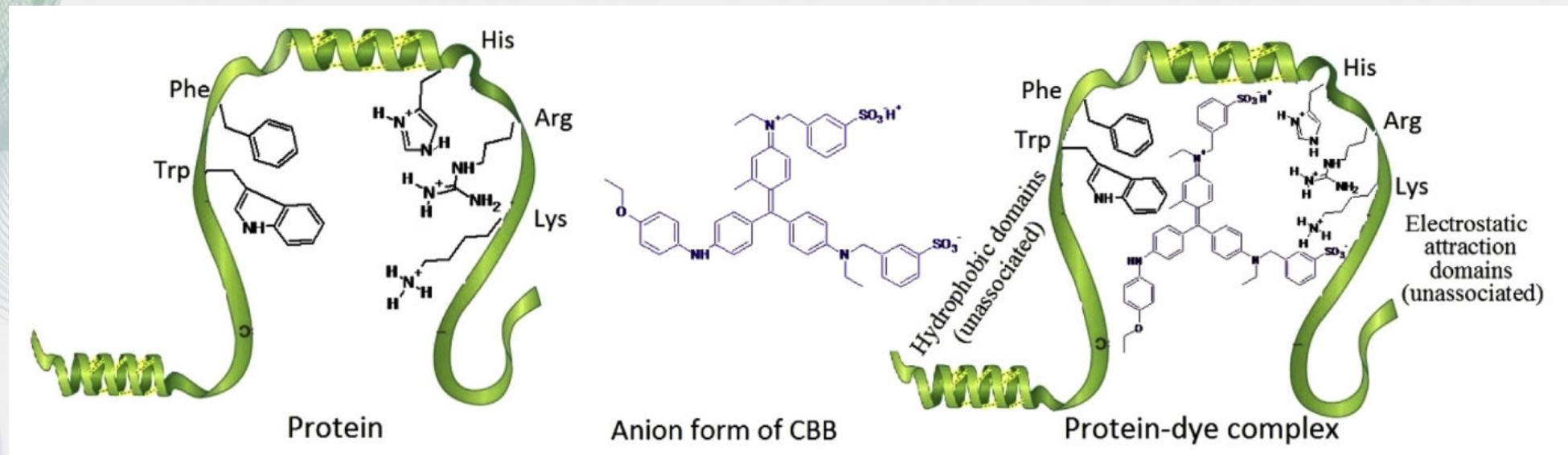
M F1 F2 F3 IEx-Peak1 IEx-Peak2



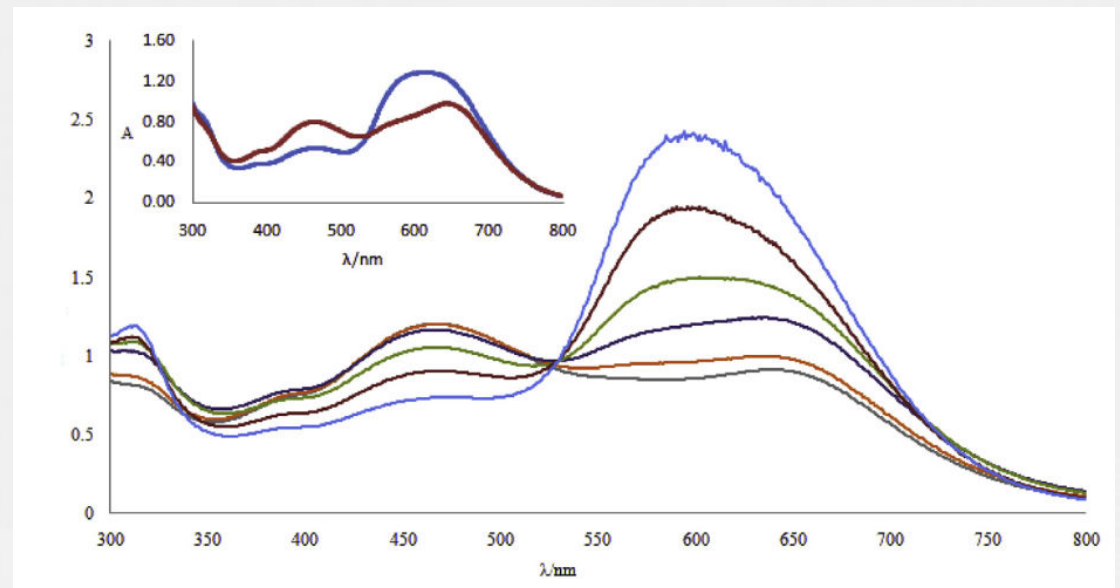
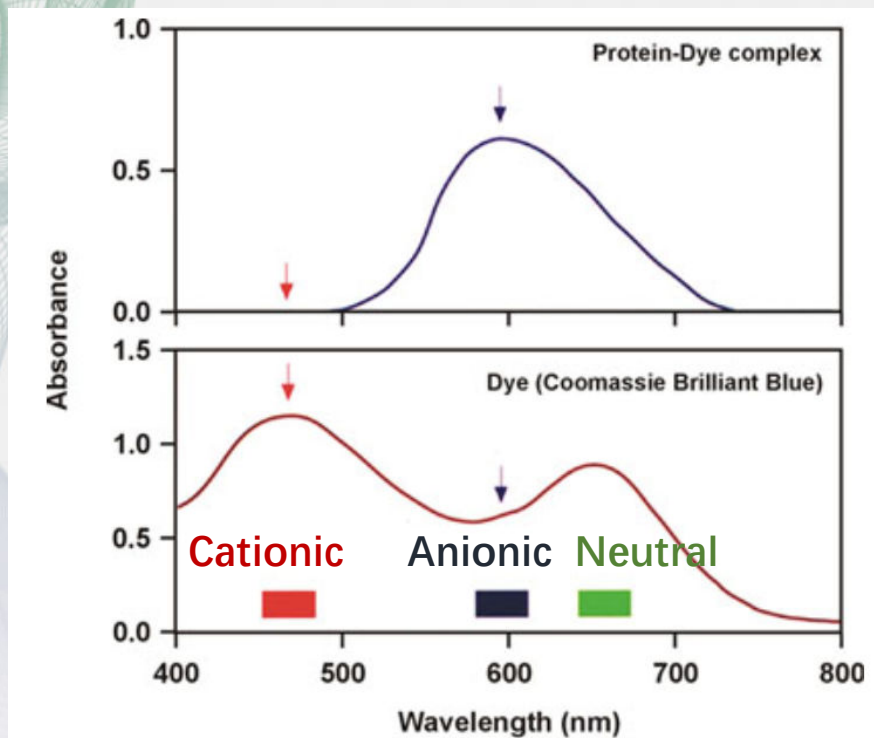


Fraction No.	Volume (mL)	Protein concentration (mg/mL)	Total protein (mg)	Specific activity [U, $\mu\text{mol}/(\text{min}\cdot\text{mg})$]	Total activity ($\mu\text{mol}/\text{min}$)	Fold purification	% Recovery

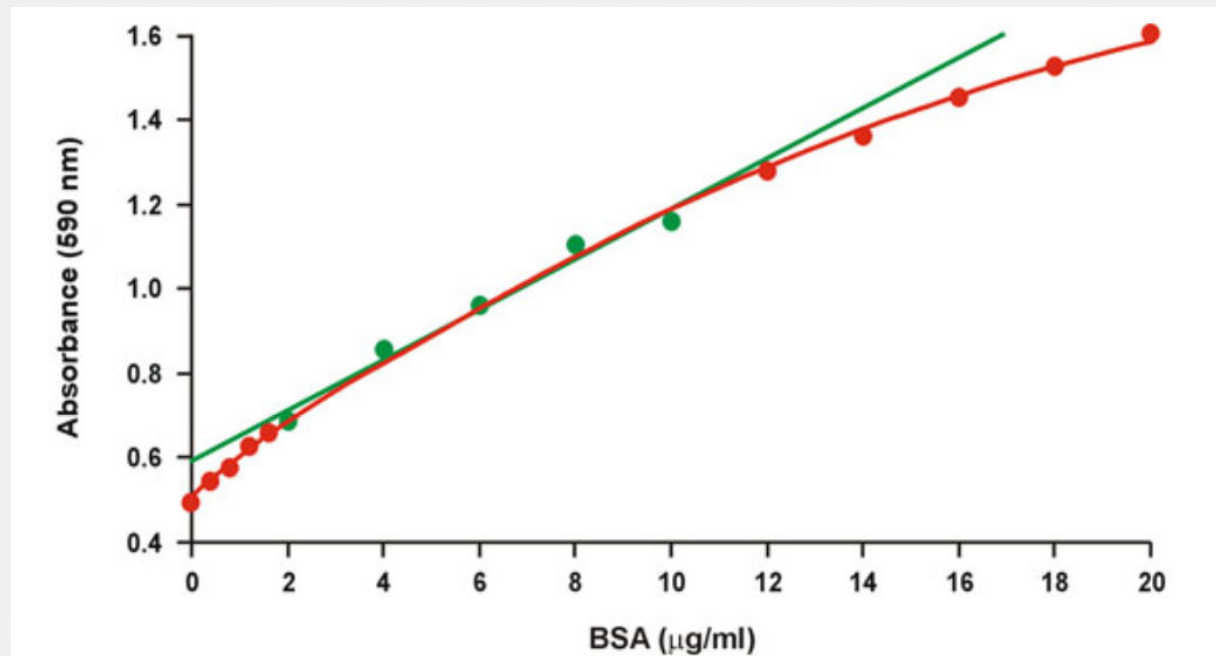
Bradford Protein Assay



Bradford Protein Assay



Bradford Protein Assay



	1	2	3	4	5
A	80 μ L PBS	60 μ L PBS + 60 μ L 1/100 dilution F1	60 μ L PBS + 60 μ L 1/100 dilution F2	60 μ L PBS + 60 μ L 1/10 dilution F3	50 μ L sample from A1
B	80 μ L PBS	60 μ L PBS + 60 μ L A2 mixture (1/200 dilution F1)	60 μ L PBS + 60 μ L A3 mixture (1/200 dilution F2)	60 μ L PBS + 60 μ L A4 mixture (1/20 dilution F3)	50 μ L sample from B1
C	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L B2 mixture (1/400 dilution F1)	60 μ L PBS + 60 μ L B3 mixture (1/400 dilution F2)	60 μ L PBS + 60 μ L B4 mixture (1/40 dilution F3)	50 μ L sample from C1
D	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L C2 mixture (1/800 dilution F1)	60 μ L PBS + 60 μ L C3 mixture (1/800 dilution F2)	60 μ L PBS + 60 μ L C4 mixture (1/80 dilution F3)	50 μ L sample from D1
E	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/200 dilution F1 from A2	50 μ L 1/200 dilution F2 from A3	50 μ L 1/20 dilution F3 from A4	50 μ L sample from E1
F	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/400 dilution F1 from B2	50 μ L 1/400 dilution F2 from B3	50 μ L 1/40 dilution F3 from B4	50 μ L sample from F1
G	80 μ L 0.1 mg/mL BSA	50 μ L 1/800 dilution F1 from C2	50 μ L 1/800 dilution F2 from C3	50 μ L 1/80 dilution F3 from C4	50 μ L sample from G1
H	80 μ L 0.1 mg/mL BSA	50 μ L 1/1600 dilution F1 from D2	50 μ L 1/1600 dilution F2 from D3	50 μ L 1/160 dilution F3 from D4	50 μ L sample from H1

Figure 1 Layout of 96-well plate for protein concentration determination

■ step 3; ■ step 5; ■ step 6; ■ step 7; ■ + ■ step 9

Serial dilution of BSA:
use ddH₂O

	1	2	3	4	5
A	80 μ L PBS	60 μ L PBS + 60 μ L 1/100 dilution F1	60 μ L PBS + 60 μ L 1/100 dilution F2	60 μ L PBS + 60 μ L 1/10 dilution F3	50 μ L sample from A1
B	80 μ L PBS	60 μ L PBS + 60 μ L A2 mixture (1/200 dilution F1)	60 μ L PBS + 60 μ L A3 mixture (1/200 dilution F2)	60 μ L PBS + 60 μ L A4 mixture (1/20 dilution F3)	50 μ L sample from B1
C	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L B2 mixture (1/400 dilution F1)	60 μ L PBS + 60 μ L B3 mixture (1/400 dilution F2)	60 μ L PBS + 60 μ L B4 mixture (1/40 dilution F3)	50 μ L sample from C1
D	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L C2 mixture (1/800 dilution F1)	60 μ L PBS + 60 μ L C3 mixture (1/800 dilution F2)	60 μ L PBS + 60 μ L C4 mixture (1/80 dilution F3)	50 μ L sample from D1
E	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/200 dilution F1 from A2	50 μ L 1/200 dilution F2 from A3	50 μ L 1/20 dilution F3 from A4	50 μ L sample from E1
F	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/400 dilution F1 from B2	50 μ L 1/400 dilution F2 from B3	50 μ L 1/40 dilution F3 from B4	50 μ L sample from F1
G	80 μ L 0.1 mg/mL BSA	50 μ L 1/800 dilution F1 from C2	50 μ L 1/800 dilution F2 from C3	50 μ L 1/80 dilution F3 from C4	50 μ L sample from G1
H	80 μ L 0.1 mg/mL BSA	50 μ L 1/1600 dilution F1 from D2	50 μ L 1/1600 dilution F2 from D3	50 μ L 1/160 dilution F3 from D4	50 μ L sample from H1

Figure 1 Layout of 96-well plate for protein concentration determination

■ step 3; ■ step 5; ■ step 6; ■ step 7; ■ + ■ step 9

Serial dilution of
sample: use ddH₂O

	1	2	3	4	5
A	80 μ L PBS	60 μ L PBS + 60 μ L 1/100 dilution F1	60 μ L PBS + 60 μ L 1/100 dilution F2	60 μ L PBS + 60 μ L 1/10 dilution F3	50 μ L sample from A1
B	80 μ L PBS	60 μ L PBS + 60 μ L A2 mixture (1/200 dilution F1)	60 μ L PBS + 60 μ L A3 mixture (1/200 dilution F2)	60 μ L PBS + 60 μ L A4 mixture (1/20 dilution F3)	50 μ L sample from B1
C	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L B2 mixture (1/400 dilution F1)	60 μ L PBS + 60 μ L B3 mixture (1/40 dilution F2)	60 μ L PBS + 60 μ L B4 mixture (1/40 dilution F3)	50 μ L sample from C1
D	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L C2 mixture (1/800 dilution F1)	60 μ L PBS + 60 μ L C3 mixture (1/80 dilution F2)	60 μ L PBS + 60 μ L C4 mixture (1/80 dilution F3)	50 μ L sample from D1
E	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/200 dilution F1 from A2	50 μ L 1/200 dilution F2 from A3	50 μ L 1/20 dilution F3 from A4	50 μ L sample from E1
F	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/400 dilution F1 from B2	50 μ L 1/400 dilution F2 from B3	50 μ L 1/40 dilution F3 from B4	50 μ L sample from F1
G	80 μ L 0.1 mg/mL BSA	50 μ L 1/800 dilution F1 from C2	50 μ L 1/800 dilution F2 from C3	50 μ L 1/80 dilution F3 from C4	50 μ L sample from G1
H	80 μ L 0.1 mg/mL BSA	50 μ L 1/1600 dilution F1 from D2	50 μ L 1/1600 dilution F2 from D3	50 μ L 1/160 dilution F3 from D4	50 μ L sample from H1

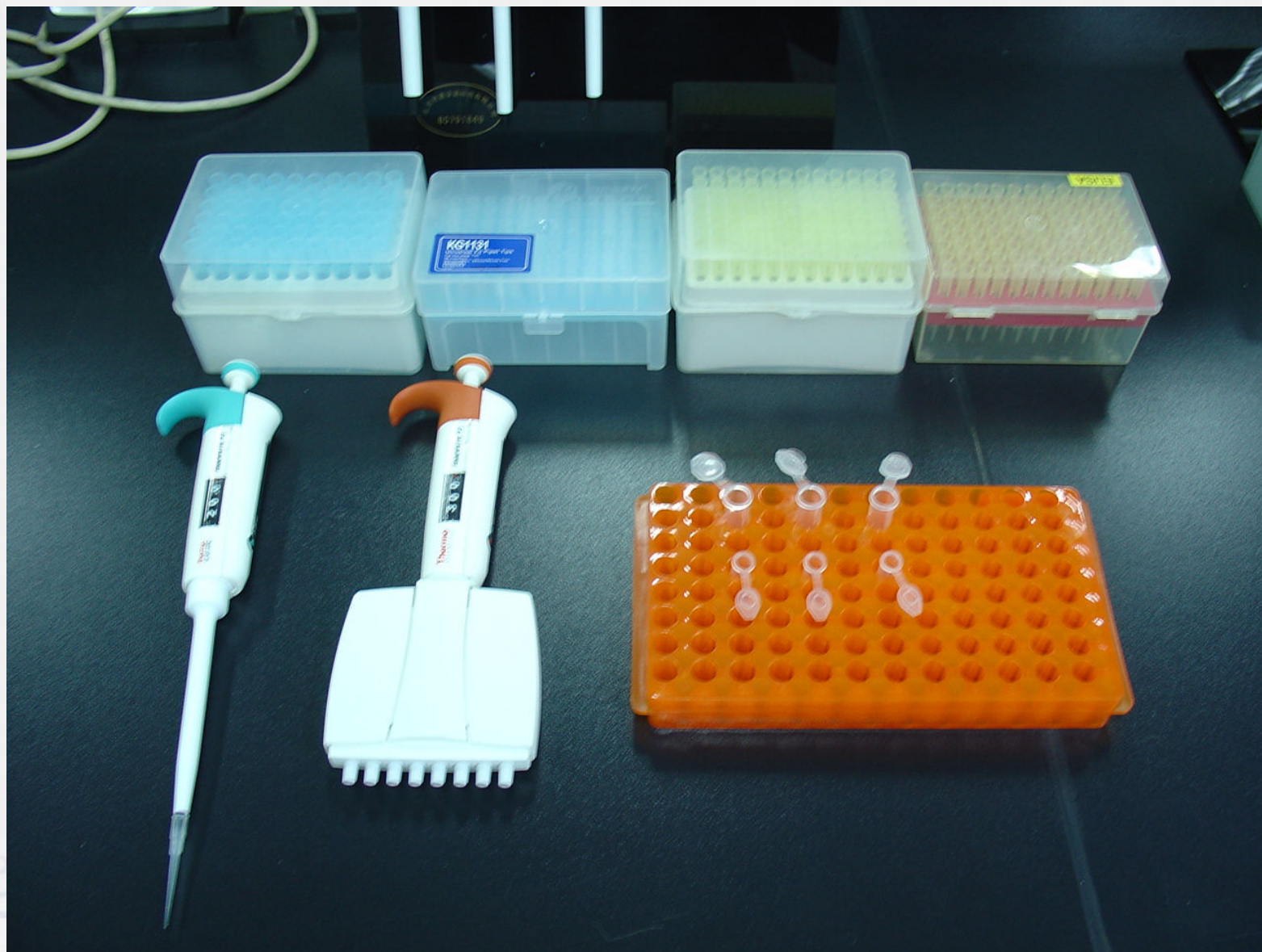
Figure 1 Layout of 96-well plate for protein concentration determination

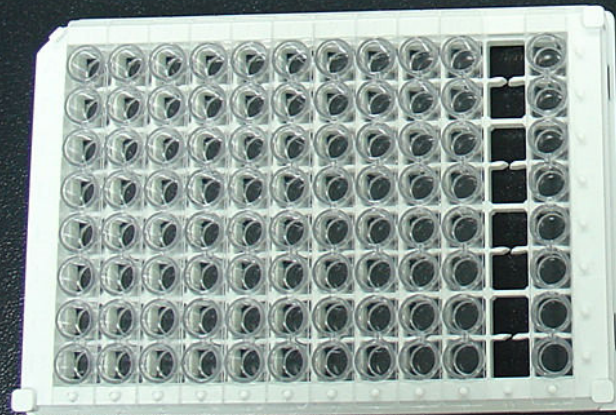
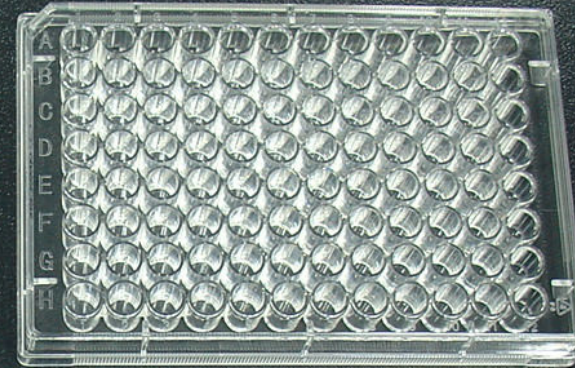
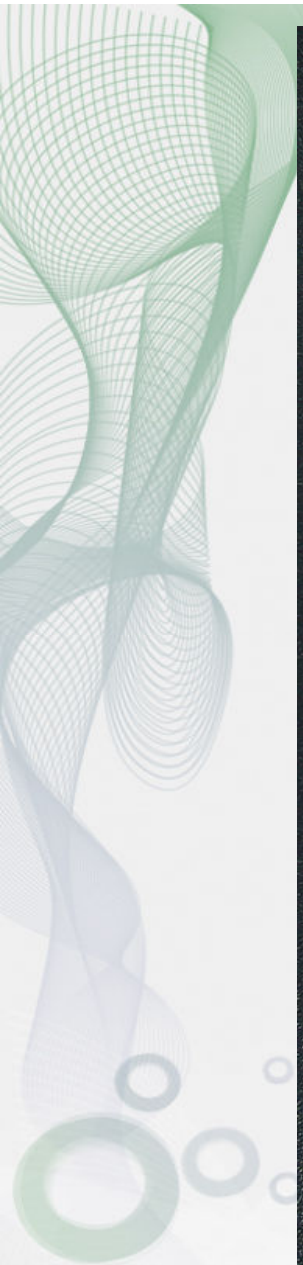
■ step 3; ■ step 5; ■ step 6; ■ step 7; ■ + ■ step 9

	1	2	3	4	5
A	80 μ L PBS	60 μ L PBS + 60 μ L 1/100 dilution F1	60 μ L PBS + 60 μ L 1/100 dilution F2	60 μ L PBS + 60 μ L 1/10 dilution F3	50 μ L sample from A1
B	80 μ L PBS	60 μ L PBS + 60 μ L A2 mixture (1/200 dilution F1)	60 μ L PBS + 60 μ L A3 mixture (1/200 dilution F2)	60 μ L PBS + 60 μ L A4 mixture (1/20 dilution F3)	50 μ L sample from B1
C	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L B2 mixture (1/400 dilution F1)	60 μ L PBS + 60 μ L B3 mixture (1/400 dilution F2)	60 μ L PBS + 60 μ L B4 mixture (1/40 dilution F3)	50 μ L sample from C1
D	60 μ L PBS + 20 μ L 0.1 mg/mL BSA	60 μ L PBS + 60 μ L C2 mixture (1/800 dilution F1)	60 μ L PBS + 60 μ L C3 mixture (1/800 dilution F2)	60 μ L PBS + 60 μ L C4 mixture (1/80 dilution F3)	50 μ L sample from D1
E	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/200 dilution F1 from A2	50 μ L 1/200 dilution F2 from A3	50 μ L 1/20 dilution F3 from A4	50 μ L sample from E1
F	40 μ L PBS + 40 μ L 0.1 mg/mL BSA	50 μ L 1/400 dilution F1 from B2	50 μ L 1/400 dilution F2 from B3	50 μ L 1/40 dilution F3 from B4	50 μ L sample from F1
G	80 μ L 0.1 mg/mL BSA	50 μ L 1/800 dilution F1 from C2	50 μ L 1/800 dilution F2 from C3	50 μ L 1/80 dilution F3 from C4	50 μ L sample from G1
H	80 μ L 0.1 mg/mL BSA	50 μ L 1/1600 dilution F1 from D2	50 μ L 1/1600 dilution F2 from D3	50 μ L 1/160 dilution F3 from D4	50 μ L sample from H1

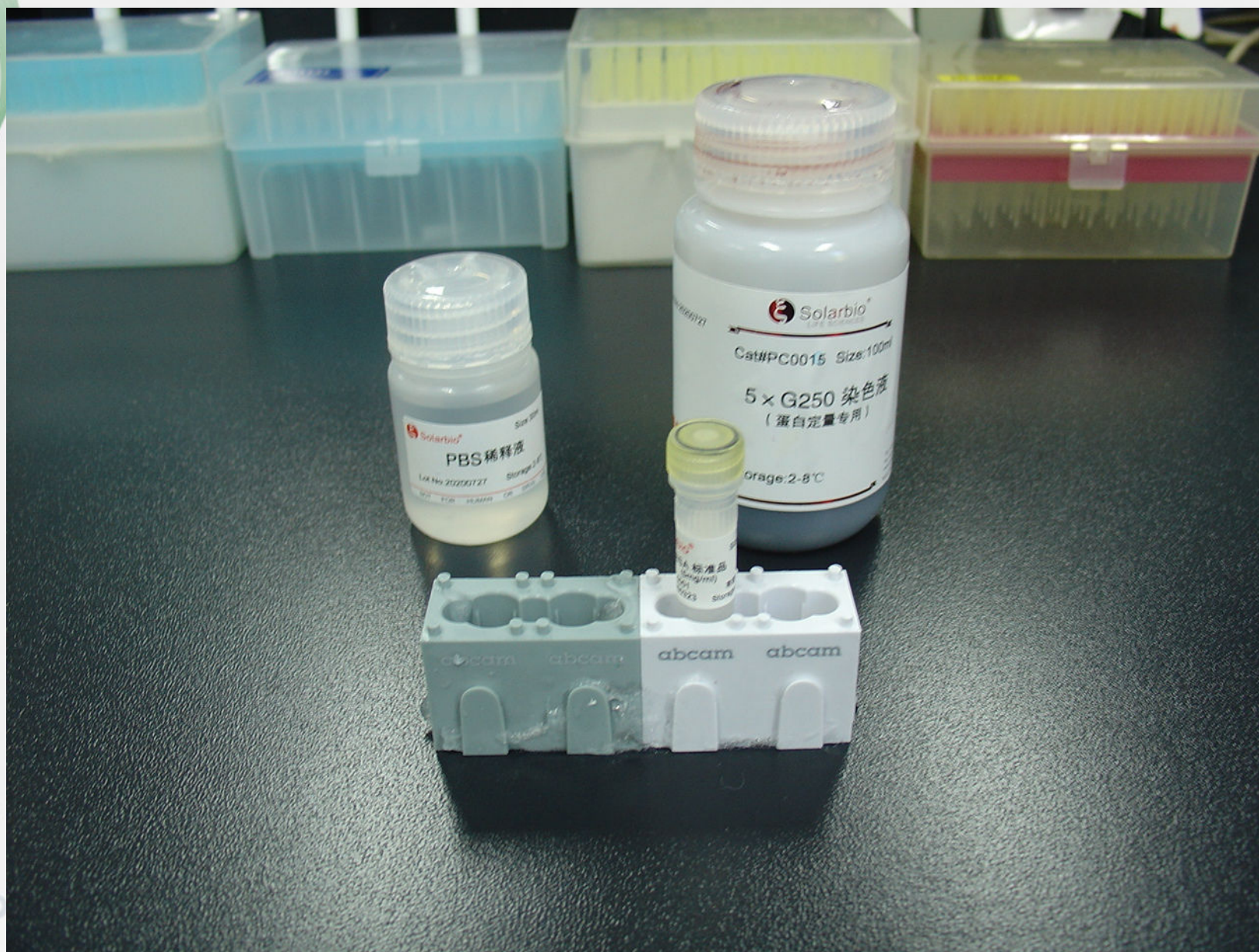
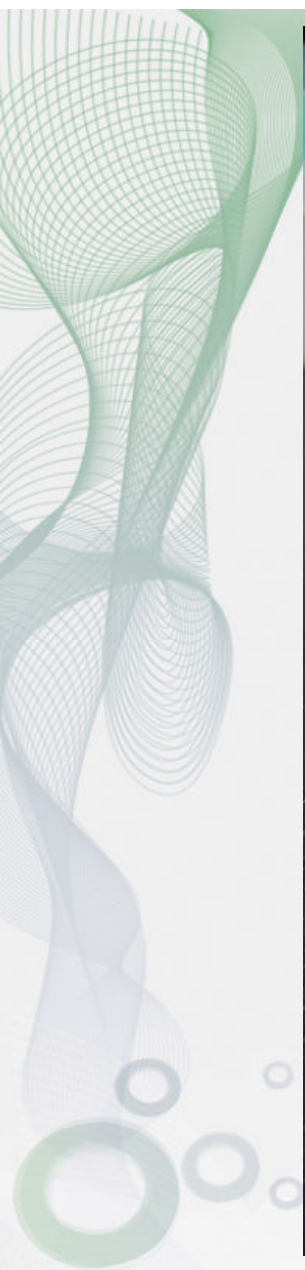
Figure 1 Layout of 96-well plate for protein concentration determination

■ step 3; ■ step 5; ■ step 6; ■ step 7; ■ + ■ step 9





Coomassie Brilliant Blue dye
dilution, use ddH₂O



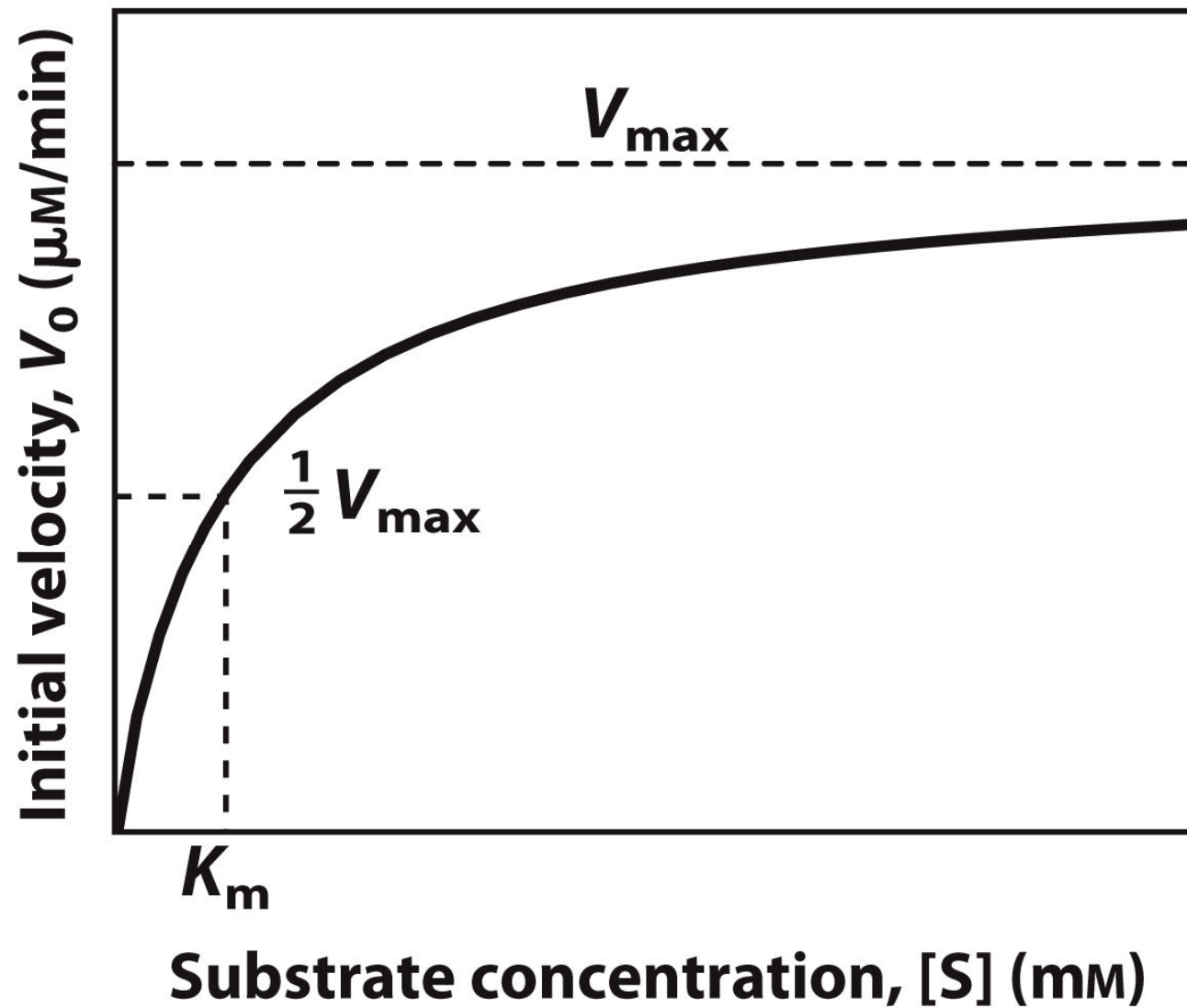


Figure 6-11
Lehninger Principles of Biochemistry, Seventh Edition
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